1. $\int \sin 3x \cdot \cos 2x \, dx$

$= \frac{1}{2} \int \sin(3x + 2x) + \sin(3x - 2x) \, dx$

$= \frac{1}{2} \int \sin(5x) + \sin x \, dx.$

$= \frac{1}{2} \times \left[-\frac{1}{5} \cos(5x) - \cos x \right] + c$

$= \frac{-\cos 5x}{10} - \frac{\cos x}{2} + c$ Ans.

2. $\int \tan^2 x \, dx$

$= \int (\sec^2 x - 1) \, dx$

$= \int \sec^2 x \, dx - \int 1 . dx$

$= \tan x - x + c$ Ans..

3. (i) Show that,

$$\int \tan x \, dx = \log |\sec x| + c$$

= put, $\cos x = z$

$$\frac{dz}{dx} = -\sin x$$

$dz = -\sin x\, dx$

$I = \int \tan x\, dx$

$= \int \frac{\sin x}{\cos x}\, dx$

$= -\int \frac{dz}{z}$

$= -\log[z] + c$

$= -\log[\cos x] + c$

$= \log\left[\frac{1}{\cos x}\right] + c$

$= \log[\sec x] + c$ [showed]

(11) $\int \cot x\, dx = \log|\sin x| + c$

= put $\sin x = z$

$\frac{dz}{dx} = \cos x$

$dz = \cos x\, dx$

Now, $\int \cot x\, dx$

$= \int \frac{\cos x}{\sin x}\, dx$

$$= \int \frac{dz}{z} dx$$

$$= \log|z| + C$$

$$= \log|\sin x| + C \text{ [Showed]}$$

4. $$\int \frac{5(x-3)^2}{x\sqrt{x}} dx$$

$$= \int \frac{5(x-3)^2}{x \cdot x^{\frac{1}{2}}} dx$$

$$= \int \frac{5(x-3)^2}{x^{\frac{3}{2}}} dx$$

$$= \int \frac{5(x^2-6x+9)}{x^{\frac{3}{2}}} dx$$

$$= 5\int \frac{x^2}{x^{\frac{3}{2}}} dx - \int \frac{6x}{x^{\frac{3}{2}}} dx + \int \frac{9}{x^{\frac{3}{2}}} dx$$

$$= 5\int x^{2-\frac{3}{2}} dx - \int 6x^{1-\frac{3}{2}} dx + \int 9x^{1-\frac{3}{2}} dx$$

$$= 5\int x^{\frac{1}{2}} dx - \int 6x^{-\frac{1}{2}} dx + \int 9x^{-\frac{3}{2}} dx$$

$$= 5\int x^{\frac{1}{2}} dx - 6\int x^{-\frac{1}{2}} dx + 9\int x^{-\frac{3}{2}} dx$$

$$= 5\left\{\frac{x^{3/2}}{3/2} - 6 \times 2x^{\frac{1}{2}} + 9(-2)x^{-\frac{1}{2}}\right\} + C$$

$$= 5 \times \frac{2}{3} x^{\frac{3}{2}} - 12x^{\frac{1}{2}} - 18x^{-\frac{1}{2}} + C$$

$$= \frac{10}{3} x^{\frac{3}{2}} - 12x^{\frac{1}{2}} - 90x^{-\frac{1}{2}} + C \text{ [Ans]}$$

5. $\cancel{\int \frac{e}{1+\tan}}$ $\int \frac{e^{\tan^{-1}x}}{1+x^2}\,dx$

= Put $z = \tan^{-1}x$

$$\frac{dz}{dx} = \frac{1}{1+x^2}$$

$$dz = \frac{1}{1+x^2}\,dx$$

Now,

$$\int e^z\,dz$$

$$= e^z + c$$

$$= e^{\tan^{-1}x} + c$$ Ans.

6. $\int x^2 (\log x)^2\,dx$

= Put $u = (\log x)^2$

$$du = 2\log x \frac{1}{x}$$

$$v = x^2\,dx$$

$$dv = \frac{x^3}{3}$$

We know,

$$\int uv = u\int v\,dx - \int \left\{\frac{du}{dx}\int v\,dx\right\}dx$$

So, $(\log x)^2 \int x^2 dx - \int \frac{d}{dx}(\log x)^2 \int x^2 dx \} dx$

$= (\log x)^2 \frac{x^3}{3} - \int \{2 \log x \frac{1}{x} \cdot \frac{x^3}{3}\} dx$

$= \frac{x^3}{3}(\log x)^2 - \frac{2}{3}\int x^2 \log x \, dx$

Now, $\int x^2 \log x \, dx$.

$U = \log x$; $V = x^2$

$= \log x \int x^2 dx - \int \{\frac{d}{dx}(\log x) \int x^2 dx\} dx$

$= \log x \frac{x^3}{3} - \int \{\frac{1}{x} \cdot \frac{x^3}{3}\} dx$

$= \frac{x^3}{3}(\log x) - \frac{1}{3}\int x^2 dx$

$= \frac{x^3}{3}(\log x) - \frac{1}{3} \cdot \frac{x^3}{3}$

$= \frac{x^3}{3}(\log x) - \frac{x^3}{9}$

Again,

$\frac{x^3}{3}(\log x)^2 - \frac{2}{3}\left[\frac{x^3}{3}(\log x) - \frac{x^3}{9}\right] + c$

$= \frac{x^3}{3}(\log x)^2 - \frac{2x^3}{9}\log x - \frac{2x^3}{27} + c$ [Ans]